

OSCEOLA, IA, USA

WMO: 720701

Lat: 41.052N

Lon: 93.689W

Elev: 1115

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 54942

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	1.5	-10.7	3.2	-2.6	-6.1	4.1	3.2	26.8	29.7	24.8	27.1	8.0	310	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	93.2	74.5	90.5	74.9	87.8	73.8	78.9	88.7	77.2	86.4	75.5	84.0	10.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	142.6	85.4	74.2	133.3	83.1	72.8	126.9	81.5	43.2	88.8	41.4	86.3	39.8	84.4	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.5	18.5	-15.2	97.1	4.6	2.5	-18.5	98.9	-21.2	100.4	-23.9	101.8	-27.2	103.7		
			-14.0	81.0	6.5	1.5	-18.7	82.1	-22.5	82.9	-26.1	83.8	-30.9	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	24.5	26.7	40.1	51.0	62.6	73.0	76.2	73.0	67.1	53.6	39.8	30.2
	DBStd	20.63	12.64	12.96	11.96	9.64	9.18	6.02	5.96	5.11	8.61	9.51	10.87	11.26
	HDD50	2937	790	657	352	104	12	0	0	0	1	72	332	617
	HDD65	5984	1255	1072	774	428	156	8	3	4	78	369	757	1080
	CDD50	3525	0	5	45	133	401	689	812	713	514	185	26	2
	CDD65	1097	0	0	2	7	81	247	350	251	141	17	1	0
	CDH74	10023	0	0	22	96	685	2348	3542	2035	1162	131	2	0
	CDH80	3685	0	0	2	14	195	879	1504	672	398	21	0	0
Wind	WSAvg	8.6	10.0	10.0	10.1	10.4	8.8	7.9	6.2	5.4	6.9	8.1	9.7	9.3
Precipitation	PrecAvg	38.1	1.2	1.5	2.2	3.9	5.2	5.4	4.5	4.2	3.7	2.4	2.0	1.4
	PrecMax	53.3	2.9	2.6	5.1	7.3	11.1	12.5	14.2	11.1	11.2	5.7	5.2	3.7
	PrecMin	26.6	0.6	0.3	0.0	1.6	1.4	0.8	1.8	0.6	1.2	0.2	0.0	0.2
	PrecStd	8.0	0.5	0.6	1.5	1.5	2.2	2.9	3.0	2.8	2.4	1.7	1.2	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	67.4	77.1	82.5	91.1	94.7	99.2	94.8	91.4	82.7	73.0	61.2
		MCWB	45.9	52.1	62.3	62.7	71.1	75.1	73.3	72.7	74.3	68.0	58.1	53.0
	2%	DB	50.2	58.0	70.2	77.0	85.6	91.2	94.7	89.6	88.7	77.7	65.6	53.7
		MCWB	43.9	50.8	56.8	61.1	69.8	74.6	75.8	75.2	73.7	63.1	55.2	47.6
	5%	DB	45.3	50.3	64.3	72.5	81.2	88.4	91.1	86.4	85.5	73.1	60.6	49.8
		MCWB	39.7	43.9	54.7	58.1	66.5	73.2	76.0	73.9	72.6	61.1	51.7	45.1
	10%	DB	41.0	44.5	58.7	68.0	77.3	85.5	88.1	83.7	81.1	69.4	55.7	45.2
		MCWB	37.3	39.6	50.7	56.1	65.3	72.5	75.1	72.6	69.7	59.7	48.1	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	56.9	63.8	66.9	74.1	79.6	81.2	80.0	77.2	71.3	60.5	55.7
		MCDB	51.6	62.7	75.7	78.2	86.0	91.1	91.4	88.7	88.0	77.4	67.7	59.3
	2%	WB	43.6	49.4	60.0	63.4	71.4	77.1	79.5	77.6	75.4	68.0	56.8	49.9
		MCDB	48.3	57.6	66.8	73.9	81.8	86.9	90.0	85.6	85.1	73.9	63.3	52.1
	5%	WB	40.5	44.5	55.3	60.6	69.5	75.3	78.0	75.7	73.7	64.8	53.2	44.8
		MCDB	45.2	49.6	62.1	70.0	78.4	85.0	87.6	83.1	81.5	70.1	59.7	48.7
	10%	WB	37.1	39.3	50.9	57.8	67.6	73.8	76.2	73.9	71.8	60.9	49.0	40.7
		MCDB	41.0	44.1	58.5	65.4	75.1	83.2	84.6	81.1	78.7	67.4	55.8	45.0
Mean Daily Temperature Range	5% DB	MDBR	17.4	17.4	18.3	20.7	19.5	19.6	19.6	19.1	20.4	20.7	18.7	15.3
		MCDBR	21.6	25.3	27.2	28.4	23.5	21.9	21.8	21.5	22.1	24.3	25.2	20.5
		MCWBR	15.7	16.9	15.6	15.5	11.0	9.3	8.5	9.8	9.5	12.6	15.2	14.6
	5% WB	MCDBR	20.7	24.7	22.4	24.6	20.1	19.5	18.6	18.9	19.2	19.3	22.5	18.1
	MCWBR	15.4	17.5	14.2	15.3	10.6	10.4	9.1	10.3	9.4	13.3	16.0	14.6	
Clear-Sky Solar Irradiance	taub	0.288	0.318	0.330	0.386	0.409	0.420	0.432	0.421	0.380	0.339	0.298	0.285	
	taud	2.428	2.345	2.399	2.260	2.264	2.274	2.237	2.286	2.352	2.463	2.510	2.459	
	Ebn at Noon	277	282	292	281	276	272	267	267	271	272	272	269	
	Edh at Noon	25	32	34	42	43	42	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	620	926	1200	1518	1790	1988	2003	1740	1476	997	682	528	
	RadStd	59	66	105	108	144	187	150	107	103	114	57	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page